

SUMMARY

Software engineer specializing in enterprise AI platforms. Designed and shipped three production agent and RAG platforms at Warner Bros. Discovery used by 11+ internal teams. Architecture lead on organization-wide documents for agent authentication, evaluation, and retrieval.

EDUCATION

- **Indian Institute of Technology, Jodhpur** Jodhpur, Rajasthan
Bachelor of Technology in Computer Science and Engineering; July 2018 – June 2022

EXPERIENCE

- **Warner Bros. Discovery** Hyderabad, India
Software Development Engineer 2 (AI Platforms) Nov 2024 - Present
 - **ACME (Agent Core for Managed Execution): Enterprise Agent Platform:**
 - Designed ACME's credentials and authentication architecture: a **four-resource model** (Team, App, Agent, Sub-agent) with separate Edit and Invocation ACLs and dual auth modes (**act.as.app** via API keys and AWS Secrets Manager; **act.as.user** via Okta OIDC), plus a **sub-agent safety property** propagating only the caller's identity through every tool call, making cross-team agent reuse safe by construction.
 - Built the **ACME Playground**, a sandboxed surface to prompt-test agents, inspect tool calls and streaming events, and validate sub-agent orchestration before publishing for org-wide reuse.
 - Refactored agent configuration off local YAML into a versioned **PostgreSQL (JSONB) registry** with a draft/published/archived lifecycle and API-driven Console Builder and Catalog, enabling **multi-tenant onboarding** without code changes.
 - Shipped **multi-model serving** via AWS Bedrock inference profiles alongside OpenAI and Anthropic, standardizing model routing and fallback for production traffic.
 - Drove ACME from **CRITICAL**-severity SAST findings to **zero** in a single sprint, hardening release posture ahead of org rollout.
 - Provisioned ACME infrastructure on the org IaC platform (Terraform/Terragrunt, Helm) and bootstrapped the **BOLT dual-language monorepo** (FastAPI + Next.js) with CI/CD across DEV, INT, STG, and PROD.
 - **Knowledge Hub: Enterprise RAG-as-a-Service:**
 - Built Knowledge Hub **end-to-end**: a **self-serve retrieval service** on **AWS OpenSearch** where teams create enterprise document indices and share document and index catalogs org-wide, so any AI app can reuse existing indices without re-ingesting; consumed by multiple internal apps and backed by an org-wide architecture document I authored.
 - Implemented ingestion from Confluence, Jira, GitHub, S3, Airflow logs, and arbitrary text, with bulk upload, team-specific chunking, and a **semantic chunking** mode that splits by meaning instead of token windows.
 - Added **multimodal embedding support** (Bedrock Titan and Anthropic Claude for text/code, a Bedrock multimodal model for images), extending access to teams whose documents are diagrams, screenshots, and scanned PDFs.
 - **Daisy: Data Analytics AI Platform:**
 - Developed Daisy, a **multi-agent AI platform** used by **11+** engineering and business teams to build domain-specific chatbots with fine-grained control over tools, data sources (SQL, Text), and models, using Vercel AI SDK, FastAPI, and LangChain.
 - Cut Daisy infra spend from **~\$700 to ~\$200 per environment** (DEV/INT/STG) and **~\$1000 to ~\$800 in PROD** by right-sizing compute/storage and decommissioning unused services.
 - Built an evaluation framework using DeepEval and custom metrics for prompt quality and LLM correctness, improving evaluation accuracy by **27%**.
 - Cut chat-session storage by **~66%** by optimizing Vercel AI SDK logging, removing redundancy and improving OpenSearch indexing and query performance.
 - Led migration of the entire Daisy platform to the org's new in-house IaC accounts during an org split: moved codebases and Lambdas off the legacy Discovery account and reprovisioned infrastructure with Terraform/Terragrunt, maintaining **zero downtime**.
 - Authored **org-wide Architecture Documents** (Evaluation Platform, Daisy Vector DB API) and ran code reviews for a team of **7 engineers**.
- **Deloitte** Hyderabad, India
Software Engineer (Analyst) July 2022 – Nov 2024
 - **Outlook add-ins and Document Archival Apps:**
 - Developed and deployed S2SACC Outlook add-in using Yeoman and MS GraphAPI, facilitating document transfer to ContentCloud via email, with role-based access based on distribution groups, serving **1000+ users**.
 - Developed S2EDS Outlook add-in used to send documents to enterprise servers from mailbox using ReactJs and Yeoman, serving **500+ users**.

SKILLS

- **Languages:** Python, C/C++, JavaScript/TypeScript, Rust, Java, Assembly, SQL
- **AI & Agent Frameworks:** LangGraph, LangChain, LangSmith, OpenAI SDK, Anthropic Claude SDK, Vercel AI SDK, DeepEval, AWS Bedrock (Titan, Claude, multimodal embeddings), TensorFlow
- **Backend & Frontend:** FastAPI, SQLAlchemy, Alembic, Next.js, React, PostgreSQL, MERN, Electron, Yeoman
- **Cloud & DevOps:** AWS (EKS, OpenSearch, Lambda, S3, API Gateway, Secrets Manager), Kubernetes, Helm, Docker, Terraform, Terragrunt, Okta OIDC, Git, Mainframe (z/OS)

ACCOLADES

- **KVPY:** Recognized as a Kishore Vaigyanik Protsahan Yojana Scholar in 2017-2018
- **Deloitte:** Honored with 3 Excellence Awards for development of the S2SACC Add-in and the OpenText Server upgrade
- **Warner Bros. Discovery:** Honored with an Excellence Award for designing and building Knowledge Hub